WRITEUP

Brave Lab

Memory Forensics| CyberDefenders

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Alias	CYB3R_B0T
Category	Brave Lab
Difficulty	Medium
Date Solved	05-11-2026
Status	Solved

1 Executive Summary

The investigation focuses on analyzing a Windows memory image to identify suspicious activities, investigate process execution, PowerShell commands, network connections, and potential indicators of compromise (IOCs). Using **Volatility 3** and other forensic tools, the objective is to reconstruct user activities, identify potential malicious behavior, and understand the security incident through memory forensics.

2 Challenge Information

Challenge Name	Brave lab
Category	Memory Forensics
Difficulty	Medium
Files Provided	67-AfricanFalls2.zip
Tools	unzip,Volatility3
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Challenge Description

The Brave challenge focuses on Windows Endpoint Forensics, where investigators analyze memory images to extract system artifacts, examine running processes, investigate network connections, and reconstruct user activities. The challenge involves using **Volatility 3, PowerShell, hexed.it** to investigate suspicious activities related to Execution, Discovery and Command and Control.

Objective

To analyze the Windows memory image, identify suspicious processes and user activities investigate potential malicious commands and network connections and extract relevant forensic artifacts to determine the attacker's actions and establish evidence of potential system compromise.

3 Reconnaissance & Initial Analysis

The initial analysis involves examining the Windows memory image to identify system information, running processes, process relationships and available forensic artifacts. Volatility 3 is used to extract essential memory information and establish a baseline for further investigation of suspicious activities.

Initial Analysis Tasks:

- Identify the Windows system profile and memory information.
- Enumerate running processes using pslist.
- Examine process hierarchy using pstree.
- Identify command-line arguments using cmdline.
- Review active network connections using netscan.
- Identify potentially suspicious processes and artifacts.

Tools: Volatility 3, PowerShell, HxD.

4 Methodology / Approach

Summarize the strategy chosen and why, before walking through execution. This is the “how I thought about it” section.

Tools Used

Tool	Purpose
Volatility 3	RAM Memory Analysis
Select-String	Filtering certain keywords/patterns from output of command or text files

Hypothesis

The Windows workstation may have experienced suspicious user activity or unauthorized execution of applications. It is hypothesized that analyzing the memory image will reveal evidence of unusual process execution, PowerShell activity, external network communications, and browser usage that may help reconstruct the user's activities and identify potential security concerns.

5 Investigation — Question by Question

Walk through the solve path in the order it happened, with enough commands and screenshots that another player could reproduce it exactly.

Q1.What time was the RAM image acquired according to the suspect system?

Executed the following Volatility 3 command to retrieve system information and identify the memory acquisition timestamp:

```
python vol.py -f a.mem  windows.info.Info
```

The system information output was examined to identify the system time recorded in the memory image.

Answers: 2021-04-30 17:52

Q2.What is the SHA256 hash value of the RAM image?

Executed the following command to calculate the SHA256 hash of the memory image and verify its integrity.

```
sha256sum a.mem
```

The command generated a unique SHA256 hash representing the RAM image.

Answers: 9db01b1e7b19a3b2113bfb65e860fffd7a1630bdf2b18613d206ebf2aa0ea172

Q3.What is the process ID of brave.exe?

Executed the following command to find the process ID of brave.exe

```
python vol.py -f a.mem windows.pslist | Select-String "brave.exe"
```

The output was filtered to locate the brave.exe process and identify its corresponding Process ID (PID)

Answers: 4856

Q4.How many established network connections were there at the time of acquisition?

Executed the following command to identify and filter established network connections from the Windows memory image.

```
python vol.py -f a.mem windows.netscan | Select-String "ESTABLISHED"
```

The output was filtered to display only network connections in the ESTABLISHED state. The total number of matching entries was counted to determine the number of established connections.

Answers: 10

Q5.Which domain name does Chrome have an established network connection with?

Executed the following Volatility 3 command to identify established network connections and extract the corresponding foreign IP addresses.

```
python vol.py -f a.mem windows.netscan |Select-String "ESTABLISHED" | Select-String "chrome.exe"
```

The extracted foreign IP address was then investigated using DNSChecker to identify its associated domain name through DNS lookup.

Answers: protonmail.ch

Q6.What is the MD5 hash value of the process executable for PID 6988?

Executed the following Volatility 3 command to identify and dump the executable associated with Process ID (PID) 6988 from the Windows memory image.

```
python vol.py -f a.mem windows.pslist --pid 6988 --dump
```

The dumped executable file was then uploaded to Hash-File Online to calculate its MD5 hash value.

Answers: 0b493d8e26f03ccd2060e0be85f430af

Q7.Can you identify the word that begins at offset 0x45BE876 and is 6 bytes long?

Opened the memory dump file using HexEd.it. Navigated to the specified offset `0x45BE876` and examined the corresponding hexadecimal and ASCII representation. The 6-byte sequence was decoded to identify the word.

The ASCII representation at the specified offset was examined to extract the 6-byte word.

Answers: hacker

Q8.What is the creation date and time of the parent process of powershell.exe?

First, executed the following command to identify **powershell.exe** in the process tree and retrieve its Parent Process ID (PPID).

```
python vol.py -f a.mem windows.pstree | Select-String "powershell.exe"
```

Next, used the identified PPID (4352) to search the process tree and examine the corresponding parent process details and creation time.

```
python vol.py -f a.mem windows.pstree | Select-String "4352"
```

The process tree output was examined to identify the parent process associated with PowerShell. The matching PPID was used to locate the corresponding process entry and retrieve its creation timestamp.

Answers: 2021-04-30 17:39

Q9.What is the full path and name of the last file opened in notepad?

Executed the following Volatility 3 command to search for `notepad.exe` and retrieve its command-line arguments from the Windows memory image.

```
python vol.py -f a.mem windows.cmdline | Select-String "notepad"
```

The output revealed that **notepad.exe** was executed with a file located in the user's temporary directory.

Answers: C:\Users\JOHNDO~1\AppData\Local\Temp\7zO4FB31F24\accountNum

Q10.How long did the suspect use Brave browser? (In Hours)

Executed the following Volatility 3 command to examine Windows UserAssist registry artifacts and identify Brave Browser execution activity.

```
python vol.py -f a.mem windows.registry.userassist | Select-String "brave"
```

The UserAssist registry artifacts from John Doe's NTUSER.DAT revealed Brave Browser execution activity. The application had an execution count of 9, a focus count of 22, and a recorded total focus time of approximately 4 hours.

Answers: 4

7 Proof of Concept

Walk through the solve path in the order it happened, with enough commands and screenshots that another player could reproduce it exactly.

Q1.What time was the RAM image acquired according to the suspect system?

```
PS C:\Users\DST\Music\volatility3> python vol.py -f a.mem  windows.info.Info
Volatility 3 Framework 2.28.3
Progress:  100.00               PDB scanning finished
Variable        Value

Kernel Base     0xf8043cc00000
DTB     0x1aa000
Symbols file:///C:/Users/DST/Music/volatility3/volatility3/symbols/windows/ntkrnlmp.pdb/769C521E4833ECF72E21F02BF33691A5-1.json.xz
Is64Bit True
IsPAE   False
layer_name      0 WindowsIntel32e
memory_layer    1 FileLayer
KdVersionBlock  0xf8043d80f368
Major/Minor     15.19041
MachineType     34404
KeNumberProcessors      4
SystemTime      2021-04-30 17:52:19+00:00
NtSystemRoot    C:\Windows
NtProductType   NtProductWinNt
NtMajorVersion  10
NtMinorVersion  0
PE MajorOperatingSystemVersion  10
PE MinorOperatingSystemVersion  0
PE Machine      34404
PE TimeDateStamp        Tue Oct 11 07:04:26 1977
```

Q2.What is the SHA256 hash value of the RAM image?



Q3.What is the process ID of brave.exe?

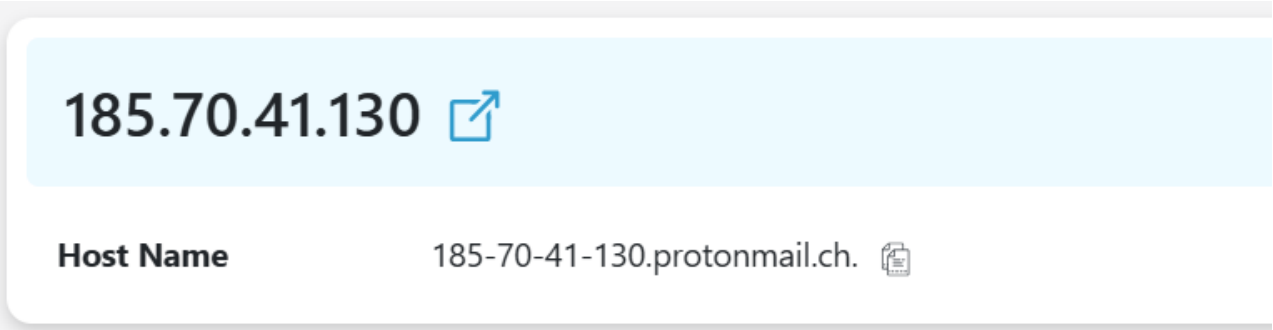
```
PS C:\Users\DST\Music\volatility3> python vol.py -f a.mem windows.pslist | Select-String "brave.exe"
Progress:  100.00               PDB scanning finished
4856    1872    brave.exe       0xbf0f6ca782c0  0       -       1       False   2021-04-30 17:48:45.000000 UTC  2021-04-30 17:50:56.0
00000 UTC       Disabled
```

Q4.How many established network connections were there at the time of acquisition?

```
PS C:\Users\DST\Music\volatility3> python vol.py -f a.mem windows.netscan |Select-String "ESTABLISHED"
Progress:  100.00               PDB scanning finished
0xbf0f6a53ca20  TCPv4   10.0.2.15       49833   52.230.222.68   443     ESTABLISHED     2812    svchost.exe     2021-04-30 17:50:07.0
00000 UTC
0xbf0f6ad16050  TCPv4   10.0.2.15       49829   142.250.191.208 443     ESTABLISHED     5624    svchost.exe     2021-04-30 17:49:58.0
00000 UTC
0xbf0f6ad1fad0  TCPv4   10.0.2.15       49847   52.230.222.68   443     ESTABLISHED     2812    svchost.exe     2021-04-30 17:52:17.0
00000 UTC
0xbf0f6c6352b0  TCPv4   10.0.2.15       49842   52.113.196.254  443     ESTABLISHED     5104    SearchApp.exe   2021-04-30 17:51:25.0
00000 UTC
0xbf0f6c7104d0  TCPv4   10.0.2.15       49778   185.70.41.130   443     ESTABLISHED     1840    chrome.exe      2021-04-30 17:45:00.0
00000 UTC
0xbf0f6cd4fa20  TCPv4   10.0.2.15       49837   204.79.197.200  443     ESTABLISHED     5104    SearchApp.exe   2021-04-30 17:51:18.0
00000 UTC
0xbf0f6d0c64a0  TCPv4   10.0.2.15       49843   204.79.197.222  443     ESTABLISHED     5104    SearchApp.exe   2021-04-30 17:51:26.0
00000 UTC
0xbf0f6d51c4a0  TCPv4   10.0.2.15       49838   13.107.3.254    443     ESTABLISHED     5104    SearchApp.exe   2021-04-30 17:51:23.0
00000 UTC
0xbf0f6d525a20  TCPv4   10.0.2.15       49845   23.101.202.202  443     ESTABLISHED     1156    MsMpEng.exe     2021-04-30 17:51:36.0
00000 UTC
0xe80000193a20  TCPv4   10.0.2.15       49845   23.101.202.202  443     ESTABLISHED     1156    MsMpEng.exe     2021-04-30 17:51:36.0
00000 UTC
```

Q5.Which domain name does Chrome have an established network connection with?

```
PS C:\Users\DST\Music\volatility3> python vol.py -f a.mem windows.netscan |Select-String "ESTABLISHED" | Select-String "chrome.exe"
Progress:  100.00               PDB scanning finished
0xbf0f6c7104d0  TCPv4   10.0.2.15       49778   185.70.41.130   443     ESTABLISHED     1840    chrome.exe      2021-04-30 17:45:00.0
00000 UTC
```

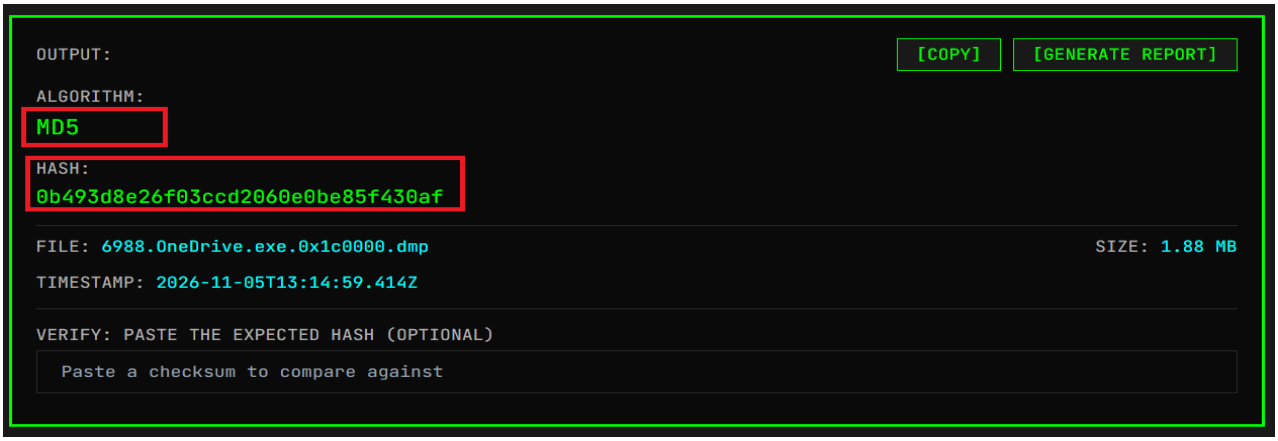


Q6.What is the MD5 hash value of the process executable for PID 6988?

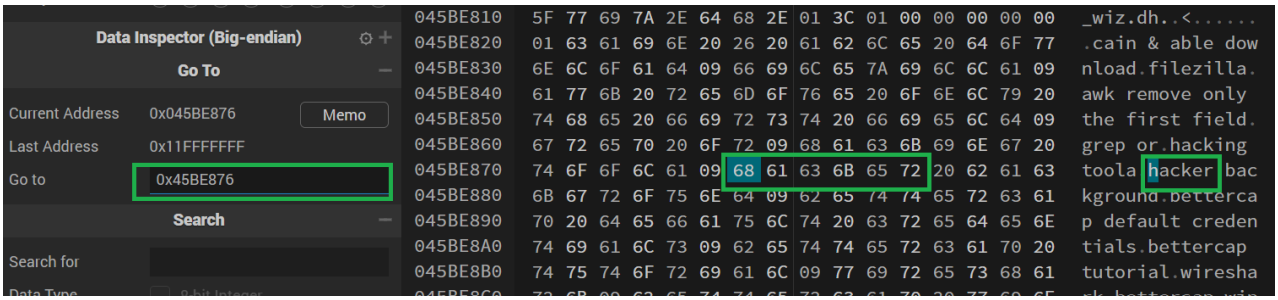
```
PS C:\Users\DST\Music\volatility3> python vol.py -f a.mem windows.pslist --pid 6988 --dump
Volatility 3 Framework 2.28.3
Progress:  100.00               PDB scanning finished
PID     PPID    ImageFileName   Offset(V)       Threads Handles SessionId       Wow64   CreateTime      ExitTime        File output

6988    4352    OneDrive.exe    0xbf0f6d4262c0  26      -       1       True    2021-04-30 17:40:01.000000 UTC  N/A     6988.OneDrive
.exe.0x1c0000.dmp
```

```
-a----          11/5/2026   7:01 PM        1974272 6988.OneDrive.exe.0x1c0000.dmp
```



Q7.What is the process ID of brave.exe?



Q8.What is the creation date and time of the parent process of powershell.exe?

```
PS C:\Users\DST\Music\volatility3> python vol.py -f a.mem windows.pstree | Select-String "powershell.exe"
Progress:  100.00               PDB scanning finished
*** 5096        4352    powershell.exe  0xbf0f6d97f2c0  12      -       1       False   2021-04-30 17:51:19.000000 UTC  N/A
\Device\HarddiskVolume2\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe"      C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
```

```
PS C:\Users\DST\Music\volatility3> python vol.py -f a.mem windows.pstree | Select-String "4352"
Progress:  100.00               PDB scanning finished
** 4352 4296    explorer.exe    0xbf0f6ca662c0  82      -       1       False   2021-04-30 17:39:48.000000 UTC  N/A
\Device\HarddiskVolume2\Windows\explorer.exe    C:\Windows\Explorer.EXE C:\Windows\Explorer.EXE
*** 6884        4352    VBoxTray.exe    0xbf0f6d186080  11      -       1       False   2021-04-30 17:40:01.000000 UTC  N/A
\Device\HarddiskVolume2\Windows\System32\VBoxTray.exe   "C:\Windows\System32\VBoxTray.exe"      C:\Windows\System32\VBoxTray.exe
*** 5096        4352    powershell.exe  0xbf0f6d97f2c0  12      -       1       False   2021-04-30 17:51:19.000000 UTC  N/A
\Device\HarddiskVolume2\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe"      C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
*** 6988        4352    OneDrive.exe    0xbf0f6d4262c0  26      -       1       True    2021-04-30 17:40:01.000000 UTC  N/A     \Devi
ce\HarddiskVolume2\Users\John
Doe\AppData\Local\Microsoft\OneDrive\OneDrive.exe       "C:\Users\John Doe\AppData\Local\Microsoft\OneDrive\OneDrive.exe" /background
C:\Users\John Doe\AppData\Local\Microsoft\OneDrive\OneDrive.exe
*** 1328        4352    chrome.exe      0xbf0f6d53e080  26      -       1       False   2021-04-30 17:44:52.000000 UTC  N/A     \Devi
ce\HarddiskVolume2\Program
Files\Google\Chrome\Application\chrome.exe      "C:\Program Files\Google\Chrome\Application\chrome.exe"          C:\Program
Files\Google\Chrome\Application\chrome.exe
*** 6772        4352    SecurityHealth  0xbf0f6d493080  1       -       1       False   2021-04-30 17:40:00.000000 UTC  N/A
\Device\HarddiskVolume2\Windows\System32\SecurityHealthSystray.exe      "C:\Windows\System32\SecurityHealthSystray.exe"
C:\Windows\System32\SecurityHealthSystray.exe
```

Q9.What is the full path and name of the last file opened in notepad?

```
PS C:\Users\DST\Music\volatility3> python vol.py -f a.mem windows.cmdline | Select-String "notepad"
Progress:  100.00               PDB scanning finished
2520    notepad.exe     "C:\Windows\system32\NOTEPAD.EXE" C:\Users\JOHNDO~1\AppData\Local\Temp\7zO4FB31F24\accountNum
```

Q10.How long did the suspect use Brave browser? (In Hours)

```
PS C:\Users\DST\Music\volatility3> python vol.py -f a.mem windows.registry.userassist  |Select-String "brave"
WARNING  volatility3.plugins.windows.registry.userassist: UserAssist key not found in \??\C:\Windows\ServiceProfiles\NetworkService\N
TUSER.DAT at 0xa80332915000
WARNING  volatility3.plugins.windows.registry.userassist: UserAssist key not found in \??\C:\Windows\ServiceProfiles\LocalService\NTU
SER.DAT at 0xa80332aab000

* 0xa80333cda000        \??\C:\Users\John Doe\ntuser.dat
ntuser.dat\SOFTWARE\Microsoft\Windows\CurrentVersion\Explorer\UserAssist\{CEBFF5CD-ACE2-4F4F-9178-9926F41749EA}\Count    2021-04-30
17:52:18.000000 UTC     Value   %ProgramFiles%\BraveSoftware\Temp\GUM20E0.tmp\BraveUpdate.exe   N/A     0       0       0:00:03.53100
0       N/A
* 0xa80333cda000        \??\C:\Users\John Doe\ntuser.dat
ntuser.dat\SOFTWARE\Microsoft\Windows\CurrentVersion\Explorer\UserAssist\{CEBFF5CD-ACE2-4F4F-9178-9926F41749EA}\Count    2021-04-30
17:52:18.000000 UTC     Value   %ProgramFiles%\BraveSoftware\Update\BraveUpdate.exe     N/A     0       1       0:00:24.797000  N/A
* 0xa80333cda000        \??\C:\Users\John Doe\ntuser.dat
ntuser.dat\SOFTWARE\Microsoft\Windows\CurrentVersion\Explorer\UserAssist\{CEBFF5CD-ACE2-4F4F-9178-9926F41749EA}\Count    2021-04-30
17:52:18.000000 UTC     Value   Brave   N/A     9       22      4:01:54.328000  2021-04-30 17:48:45.000000 UTC
* 0xa80333cda000        \??\C:\Users\John Doe\ntuser.dat
ntuser.dat\SOFTWARE\Microsoft\Windows\CurrentVersion\Explorer\UserAssist\{F4E57C4B-2036-45F0-A9AB-443BCFE33D9F}\Count    2021-04-30
17:51:18.000000 UTC     Value   C:\Users\Public\Desktop\Brave.lnk        N/A     8       0       0:00:00.508000  2021-04-30 17:48:45.0
00000 UTC
```

9 Lessons Learned & Remediation

Key Takeaways

- Learned how to use Volatility 3 to investigate Windows memory images and extract valuable forensic artifacts.
- Understood how to analyze process trees, command-line arguments, network connections, and UserAssist registry entries.
- Learned to identify application execution history and user activity through Windows registry artifacts.
- Gained practical experience using HexEd.it to inspect memory offsets and Hash-File Online to calculate file hashes.
- Improved the ability to correlate multiple forensic artifacts to reconstruct system activity.

Remediation (if applicable to a real-world parallel)

1. Process Monitoring: Implement endpoint monitoring to detect suspicious process execution and unusual parent-child process relationships.
2. Network Security: Monitor established network connections and investigate communication with suspicious external IP addresses or domains.
3. Endpoint Protection: Use updated antivirus/EDR solutions and application control to prevent unauthorized executable activity.
4. Audit Logging: Enable Windows process creation auditing, PowerShell logging, and registry activity monitoring.
5. Incident Response: Preserve memory dumps and relevant forensic evidence to support future investigations and incident analysis.

10 References & Appendix

10.1 References

- CyberDefenders – Brave Challenge — Challenge overview and investigation scenario.
- Volatility 3 Documentation — Documentation for memory-forensics plugins and framework usage.
- HexEd.it — Online hex editor for memory dump inspection.
- DNSChecker — DNS lookup and domain investigation.
- Hash-File Online — Online file hash calculation and analysis.

10.2 Appendix

The following supplementary materials support the investigation and provide evidence for the findings:

- windows.info output showing system information.
- windows.cmdline output showing command-line arguments.
- windows.registry.userassist output showing Brave Browser activity.
- SHA256 hash of the original memory image.
- MD5 hash of the dumped executable (PID 6988).
- HexEd.it screenshot showing the identified word at offset 0x45BE876.
- DNS lookup results for the identified foreign IP address.

End of Writeup